IGCSE: Probability

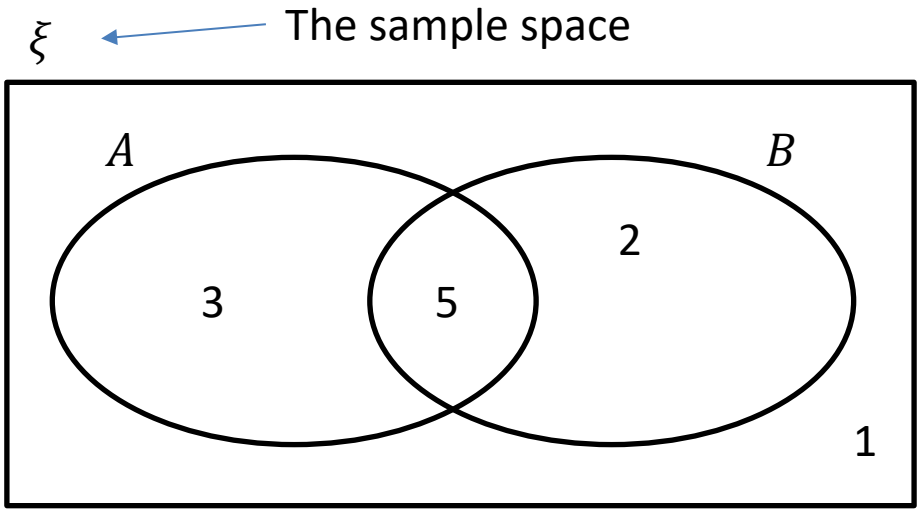
RECAP: Events

Examples of events:
Throwing a 6, throwing an odd number, tossing a heads, a randomly chosen person having a height above 1.5m.

- The sample space is the ?
- An event is ?

$$P(A) = \frac{1}{3}$$

We often use capital letters to represent an event, then use $P(A)$ to mean the probability of it.



Independent Events

When a fair coin is thrown, what's the probability of:

$P(H) =$?

And when 3 fair coins are thrown:

p(1st coin H and 2nd coin H and 3rd coin H) = ?

Therefore in this particular case we found the following relationship between these probabilities:

P(event$_1$ **and** event$_2$ **and** event$_3$)

= ?

Mutually Exclusive Events

If A and B are mutually exclusive events, they can't happen at the same time. Then:

$P(A \text{ } \underline{\textbf{or}} \text{ } B) = P(A) + P(B)$

Independent Events

If A and B are independent events, then the outcome of one doesn't affect the other. Then:

$P(A \text{ } \underline{\textbf{and}} \text{ } B) = P(A) \times P(B)$

But be careful…

1 2 3 4 5 6 7 8

P(num divisible by 2) = ?

P(num divisible by 4) = ?

P(num divisible by 2 **and** by 4) = ?

Why would it have been wrong to multiply the probabilities?

Relevant exam style question…

Dave and Bob both come into school by bus from Hounslow.
The probability that Dave is late to school is 0.7.
The probability that Bob is late to school is 0.4.

Sheila claims that the probability Dave is late to school **and** Bob is late to school is $0.7 \times 0.4 = 0.28$
Sheila is wrong. Explain why this might be.

?

Add or multiply probabilities?

Getting a 6 on a die and a T on a coin.

+ ×

Hitting a bullseye or a triple 20.

+ ×

Getting a HHT or a THT after three throws of an unfair coin (presuming we've already worked out P(HHT) and P(THT).

+ ×

Getting 3 on the first throw of a die and a 4 on the second.

+ ×

Bart's favourite colour being red and Pablo's being blue.

+ ×

Shaan's favourite colour being red or blue.

+ ×

Independent?

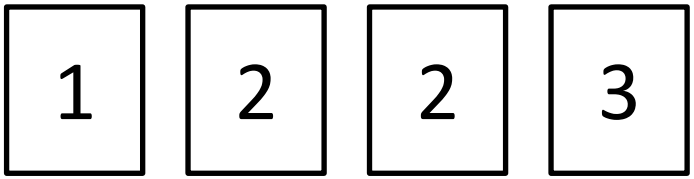
Event 1	Event 2		
Throwing a heads on the first flip.	Throwing a heads on the second flip.	No	Yes
It rains tomorrow.	It rains the day after.	No	Yes
That I will choose maths at A Level.	That I will choose Physics at A Level.	No	Yes
Have a garden gnome.	Being called Bart.	No	Yes

Test Your Understanding

a The probability that Kyle picks his nose today is 0.9. The probability that he independently eats cabbage in the canteen today is 0.3. What's the probability that Kyle picks his nose, but doesn't eat cabbage?

?

b I pick two cards from the following. What is the probability the first number is a 1 and the second number a 2?



?

c I throw 100 dice and 50 coins. What's the probability I get all sixes and all heads?

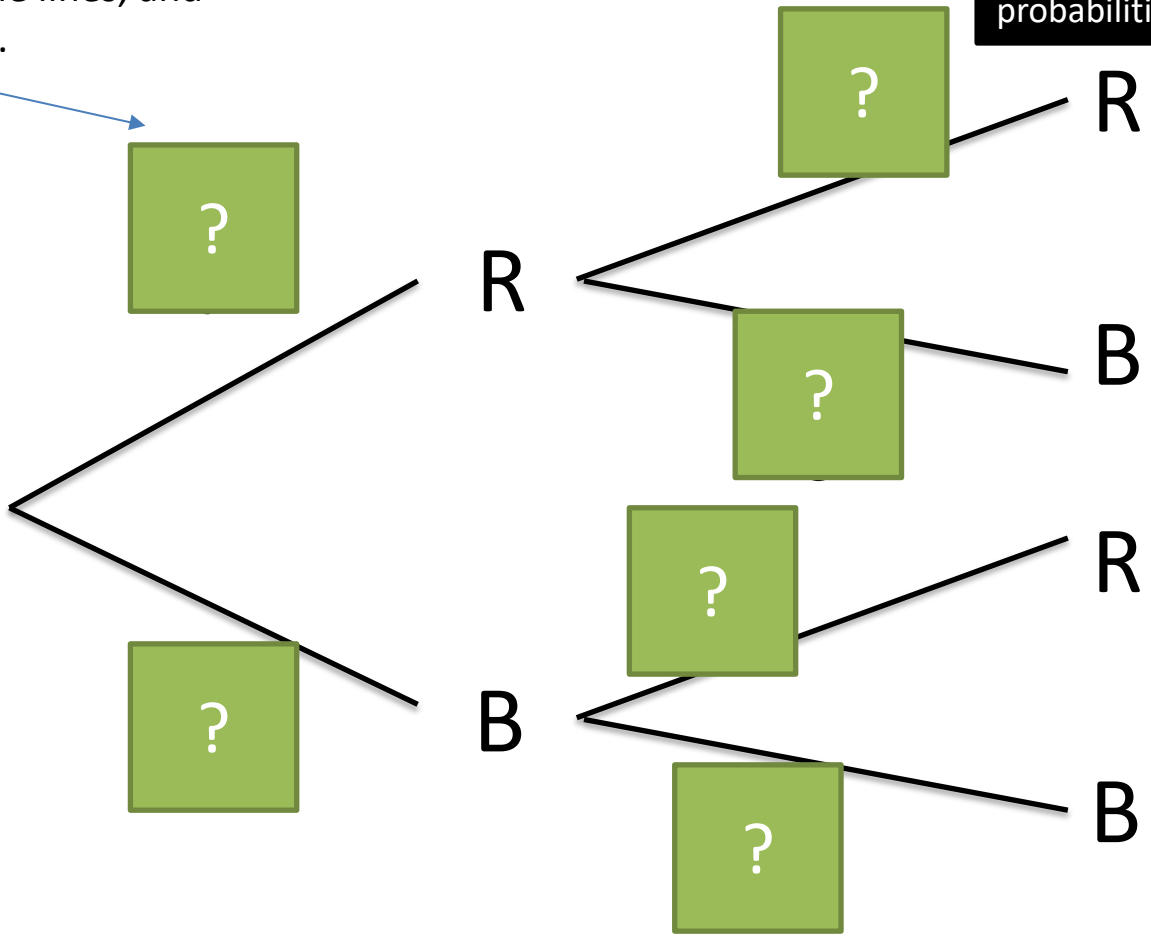
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Tree Diagrams

Question: Given there's 5 red balls and 2 blue balls. What's the probability that after two picks we have **a red ball and a blue ball**?

Tip: Note that probabilities generally go on the lines, and events at the end.

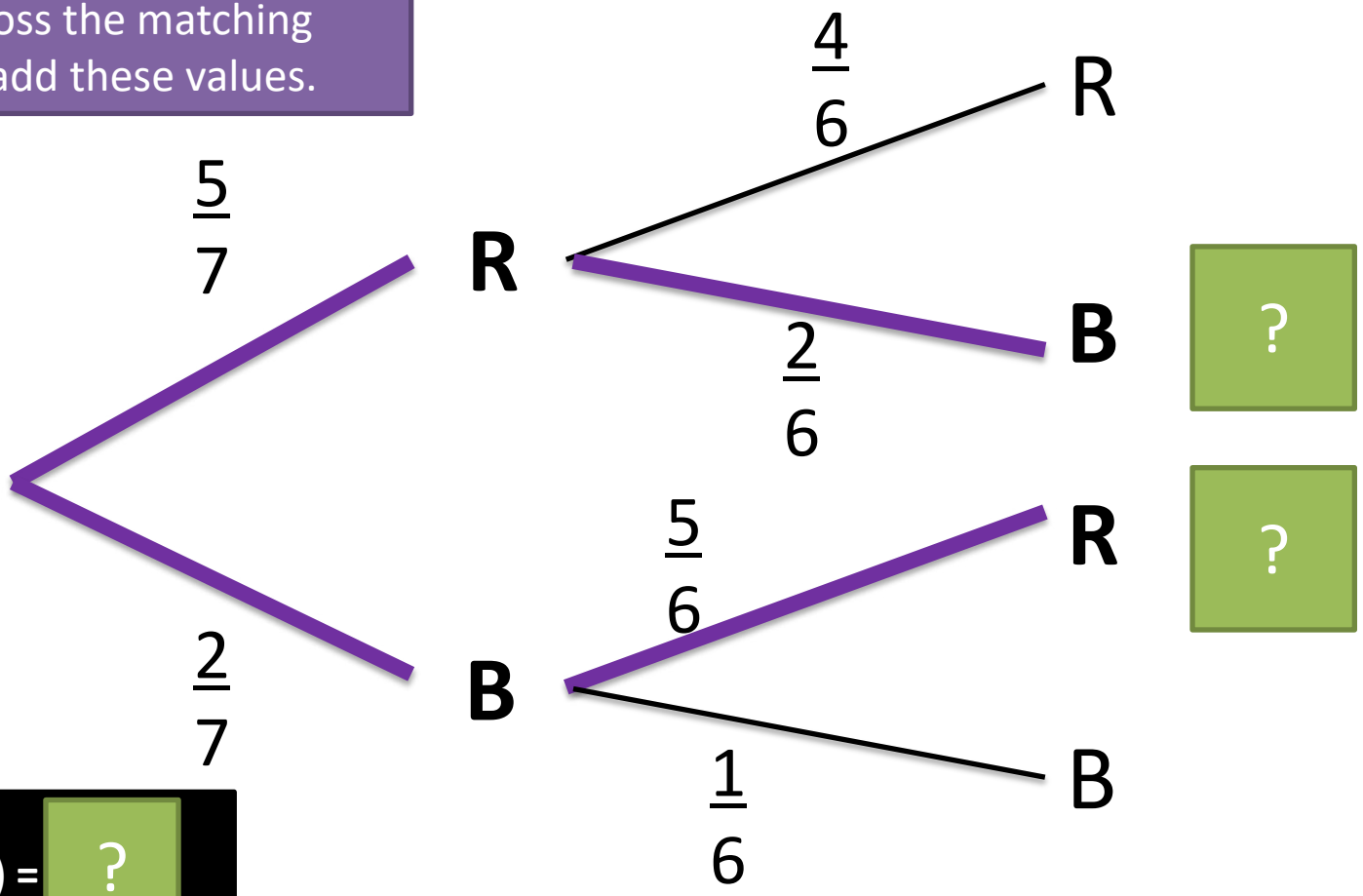
After first pick, there's less balls to choose from, so probabilities change.



Tree Diagrams

Question: Give there's 5 red balls and 2 blue balls. What's the probability that after two picks we have **a red ball and a blue ball**?

We multiply across the matching branches, then add these values.



P(red and blue) = ?

Summary

...with replacement:
The item is returned before another is chosen.
The probability of each event on each trial is fixed.

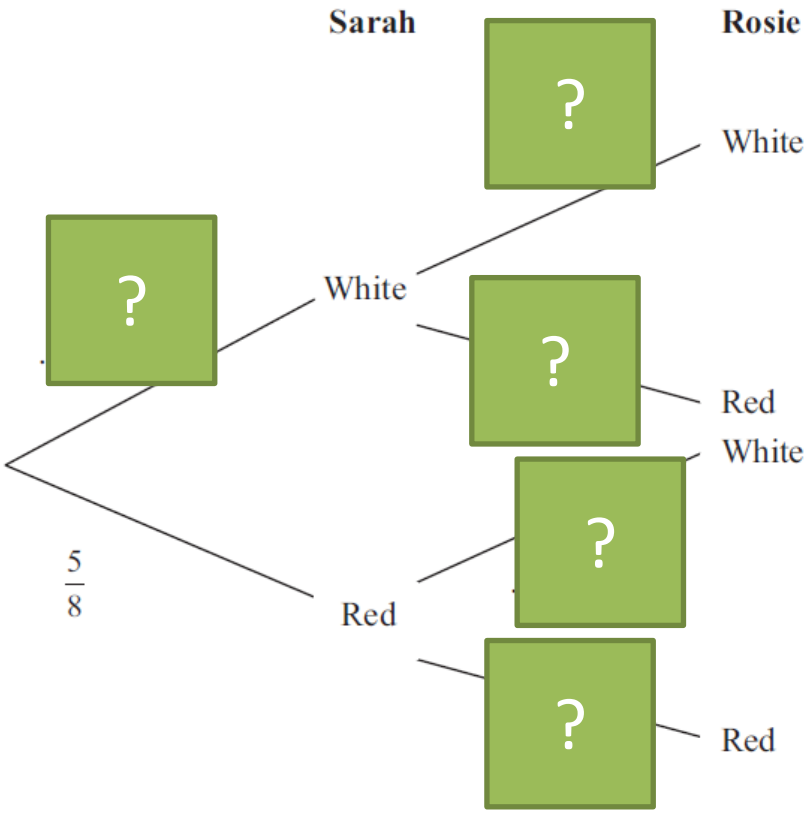
...without replacement:
The item is not returned.
- Total items decreases by 1 each time.
- Number of items of this type decreases by 1.

Note that if the question doesn't specify which, e.g. "You pick two balls from a bag", then PRESUME WITHOUT REPLACEMENT.

Example (on your sheet)

A bag contains 5 red counters and 3 white counters.
Sarah takes a counter at random from the bag. She notes its colour and replaces it.
Rosie then takes a counter at random from the bag.

(a) Complete the probability tree diagram shown below.



(b) Work out the probability that both Sarah and Rosie take a red counter.

(c) Work out the probability that only one of the counters taken is red.

?

?

Algebraic Probability Questions

There are n sweets in a bag.
6 of the sweets are orange.
The rest of the sweets are yellow.

Hannah takes at random a sweet from the bag.
She eats the sweet.

Hannah then takes at random another sweet from the bag.
She eats the sweet.

The probability that Hannah eats two orange sweets is $\frac{1}{3}$

(a) Show that $n^2 - n - 90 = 0$

This question famously made national news…

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b) Solve $n^2 - n - 90 = 0$ to find the value of n.

?

Test Your Understanding

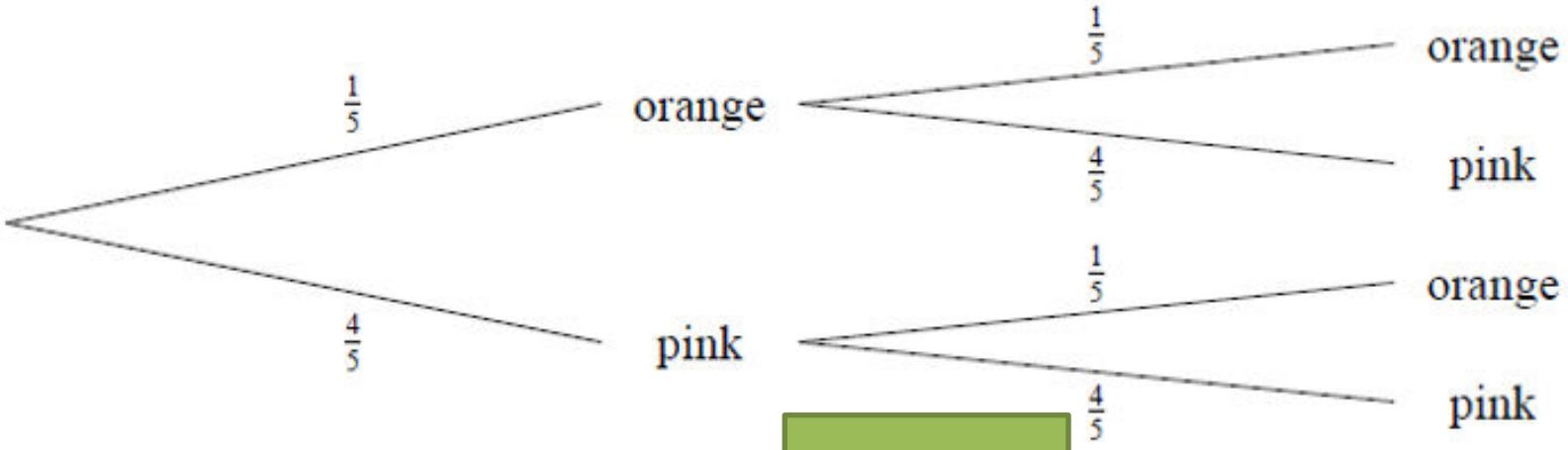
I have 5 sweets, n of which are blue. I take a sweet, eat it, then take another.

The probability both of my sweets are blue is $\frac{3}{10}$

Use the information to form an appropriate quadratic equation $n^2 + an + b = 0$, specifying the constants a and b.

$a =$? , $b =$?

Question 1

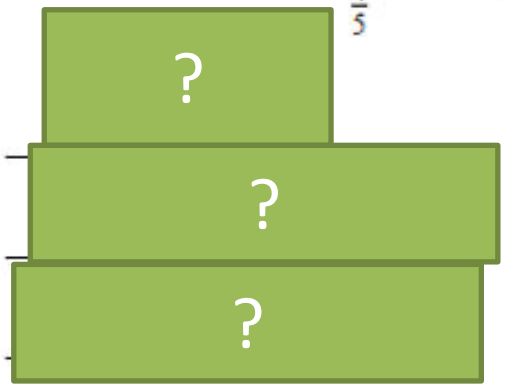


What's the probability of getting:

a) Orange twice?

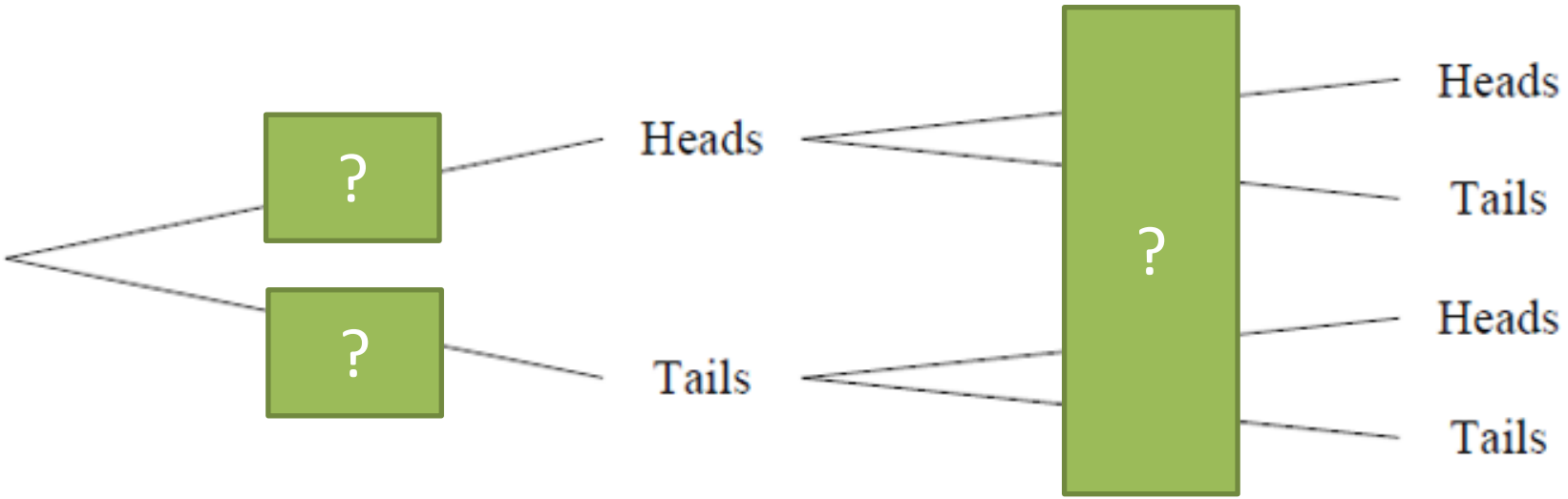
b) Seeing different colours?

c) Seeing the same colour twice?



Question 2

The probability of getting Heads on a biased coin is 0.9.



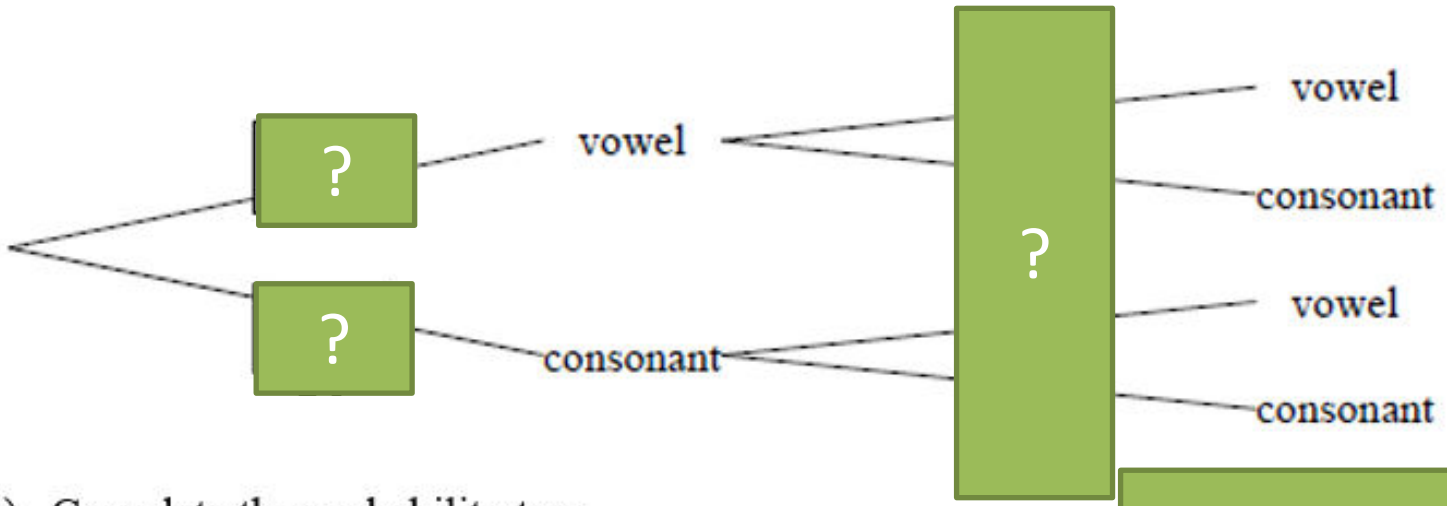
a) Complete the tree diagram.

b) What is the probability of getting Heads twice? ?

c) What is the probability of seeing one each of Heads and Tails? ?

Question 3

A bag contains the letters of 'TRANSFORMATION'. I pick two letters from the bag without replacement.

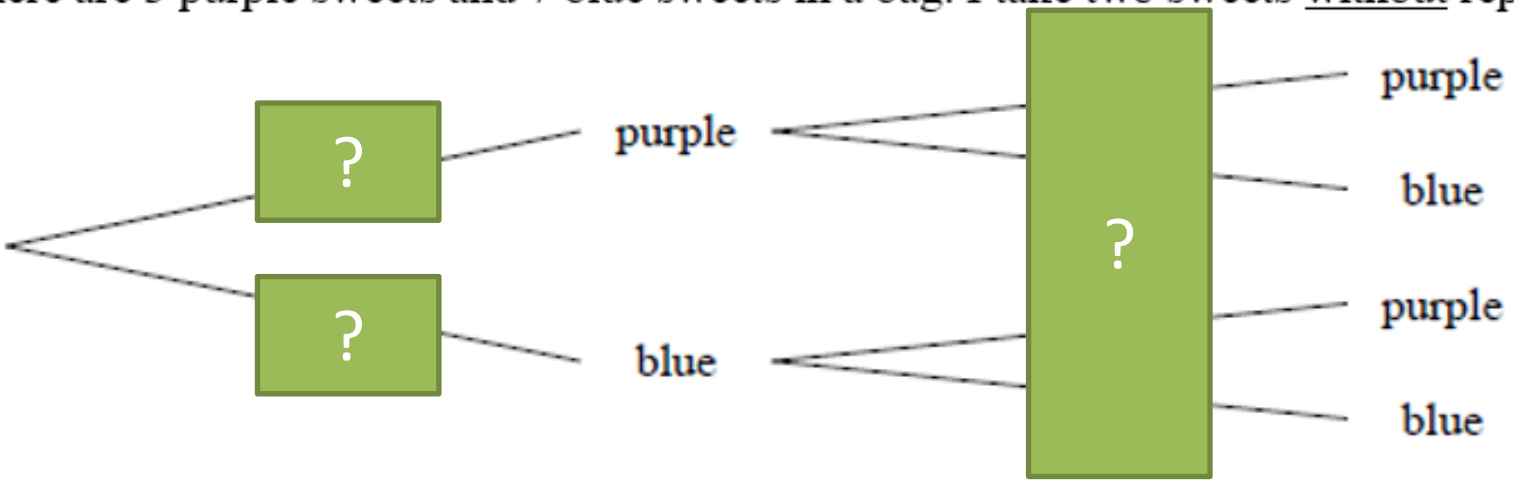


a) Complete the probability tree.
b) What is the probability of getting two consonants?

c) What is the probability of getting exactly one vowel?

Question 4

There are 3 purple sweets and 7 blue sweets in a bag. I take two sweets without replacement.



a) Complete the probability tree.

b) What is the probability that I get two sweets of different colours?

a) What is the probability that I get two sweets of the same colour?

?

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Question 5 – "The Birthday Paradox"

You wish to find the probability that in a classroom of 30 people, at least two share the same birthday.

a. It is easier to work out the probability first that no one shares the same birthday. Presuming a year is 365 days, what is the probability that the second person in the room doesn't share a birthday with the first?

?

b. What is the probability that the third person doesn't share a birthday with either of the first two?

?

c. Hence work out the probability that in the classroom, no one shares a birthday.

?

d. Hence work out the probability that at least two people share a birthday.

?

Question 6

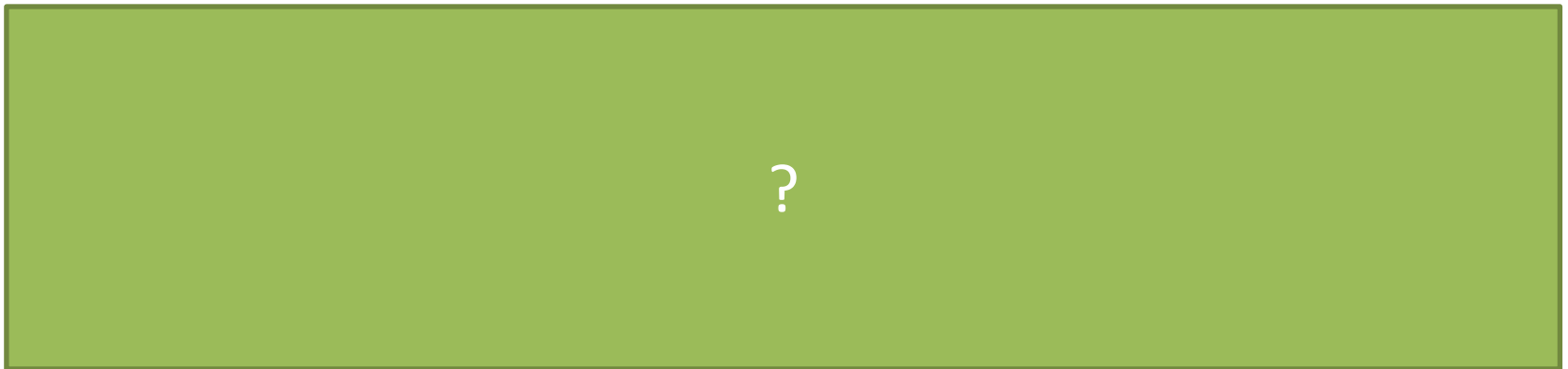
22. **[Nov 2011 NonCalc]** Jan has two boxes.
There are 6 black and 4 white counters in box A.
There are 7 black and 3 white counters in box B.

Jan takes at random a counter from box A and puts it in box B.
She then takes at random a counter from box B and puts it in box A.

(a) Complete the probability tree diagram.

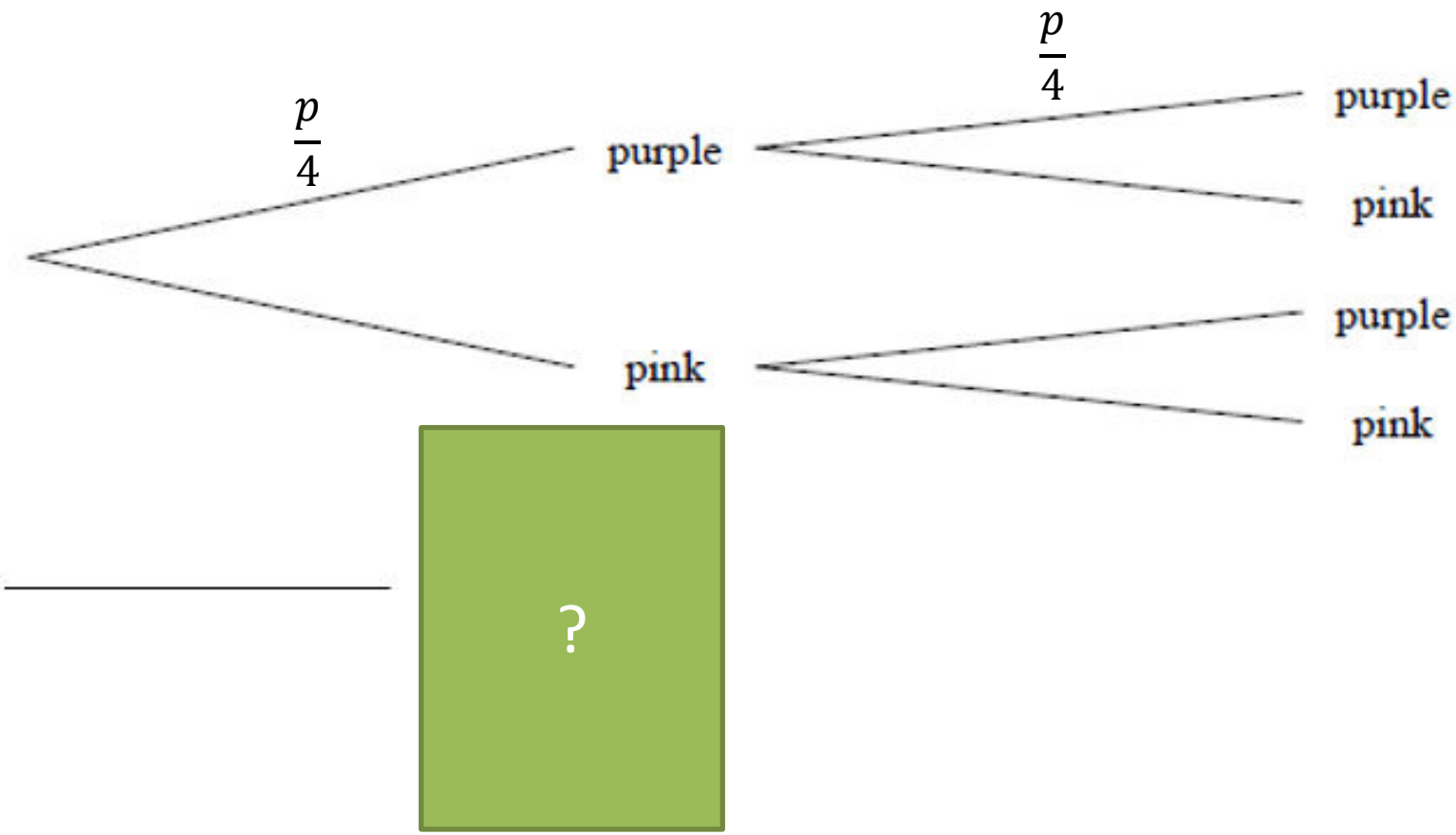


(b) Find the probability that after Jan has put the counter from box B into box A there will still be 6 black counters and 4 white counters in box A.



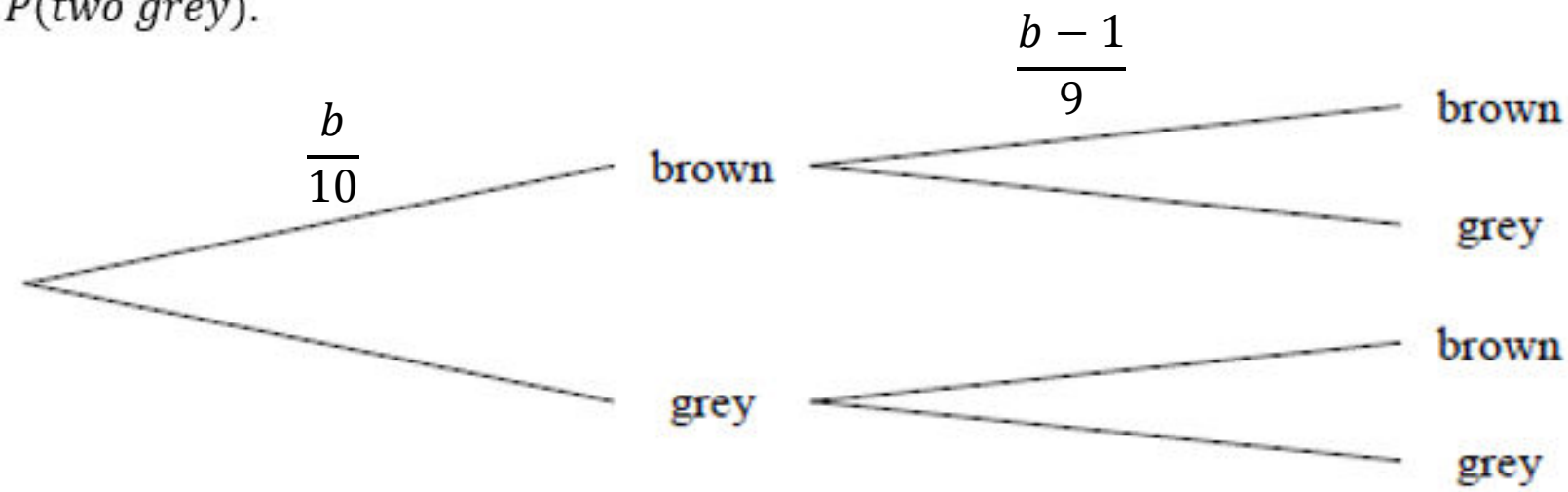
Question 7 (Algebraic Trees)

A box contains a total of 4 balls; some are purple and the rest pink. A ball is chosen at random, replaced before choosing again. If $P(two\ purple) = \frac{1}{16}$ then determine the number of purple balls.



Question 8

A bag contains a total of 10 sweets; some are brown and the rest are grey. A sweet is chosen at random and not replaced before choosing another one. If $P(two\ brown) = \frac{28}{45}$ then find $P(two\ grey)$.



?

Question ☠$_1$

[Maclaurin M68] I have 44 socks in my drawer, each either red or black. In the dark I randomly pick two socks, and the probability that they do not match is $\frac{192}{473}$. How many of the 44 socks are red?

?

Question ☠$_2$

[Maclaurin 2013 Q4] Two coins are biased in such a way that, when they are both tossed once:
(i) the probability of getting two heads is the same as the probability of getting two tails;
(ii) the probability of getting one head and one tail is $\frac{5}{8}$.
For each coin, what is the probability of getting a head?

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